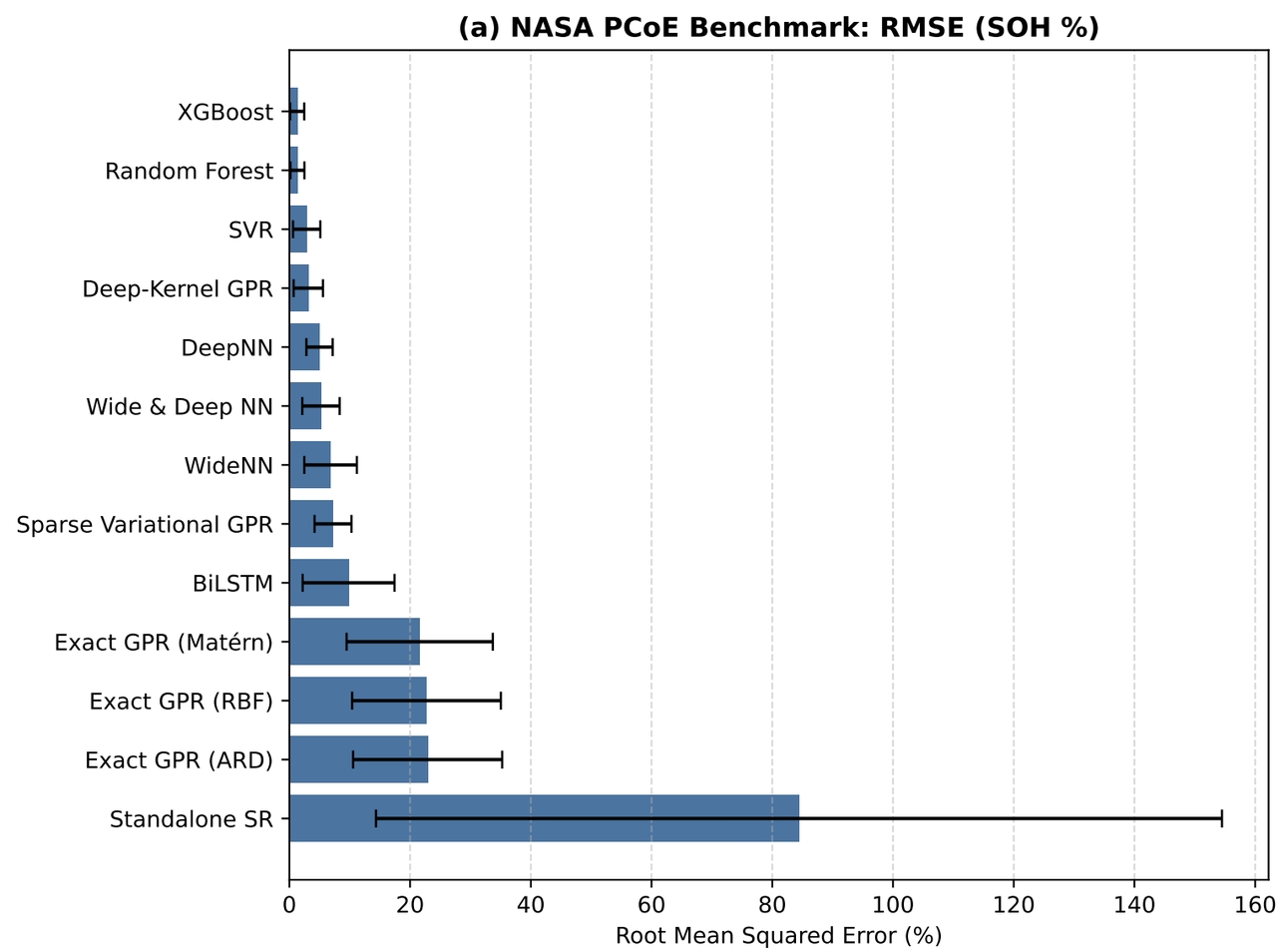
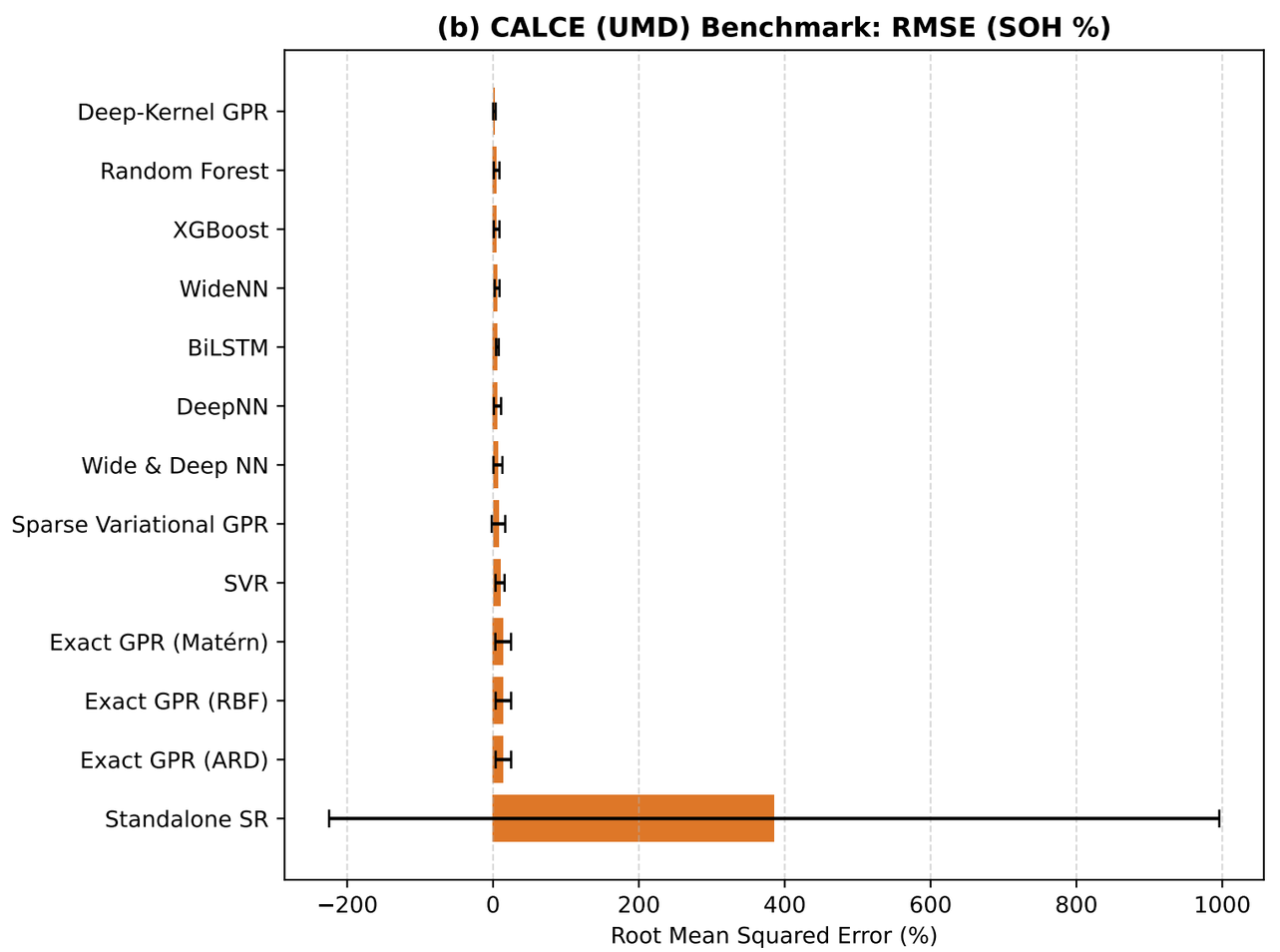
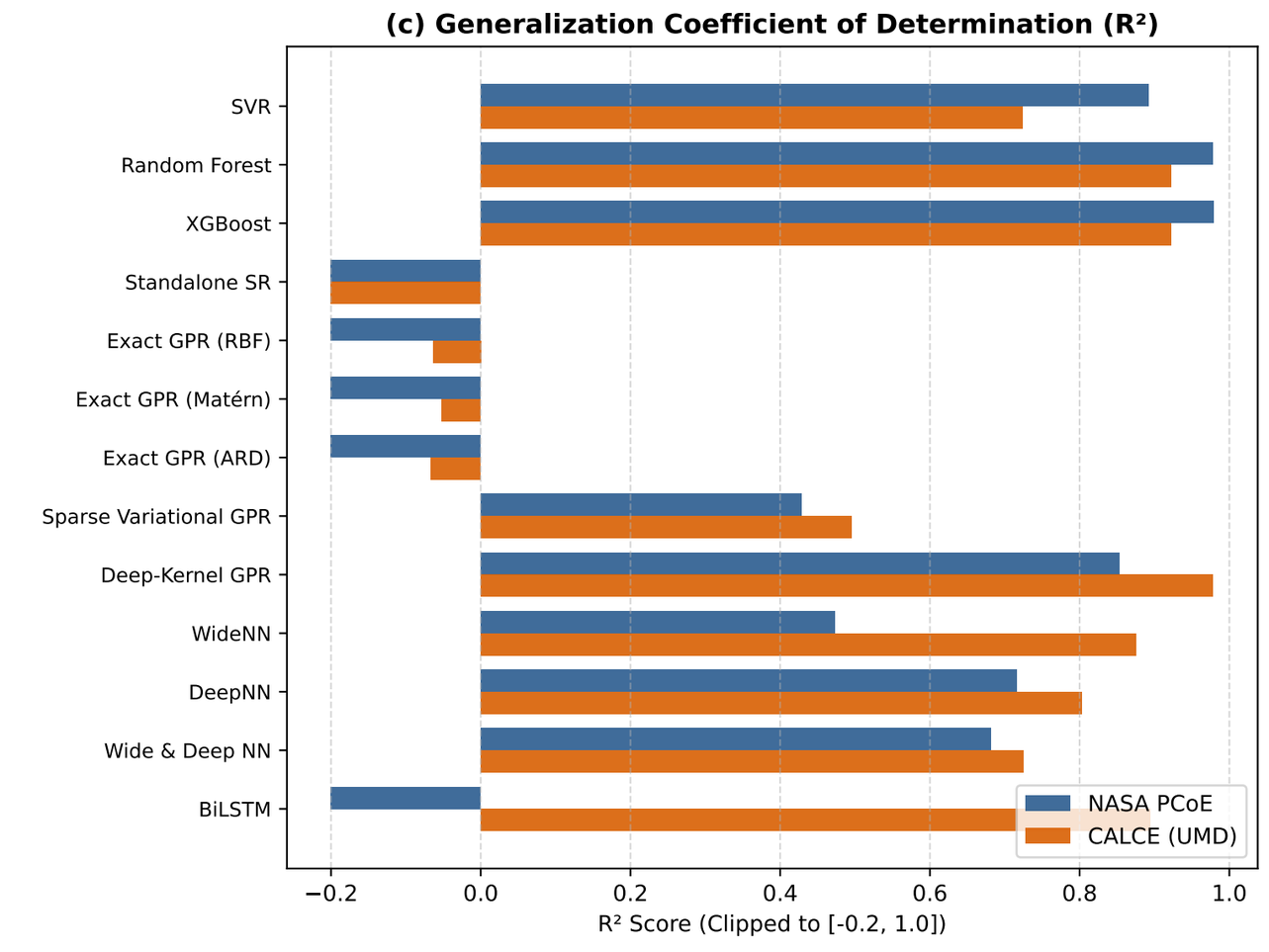
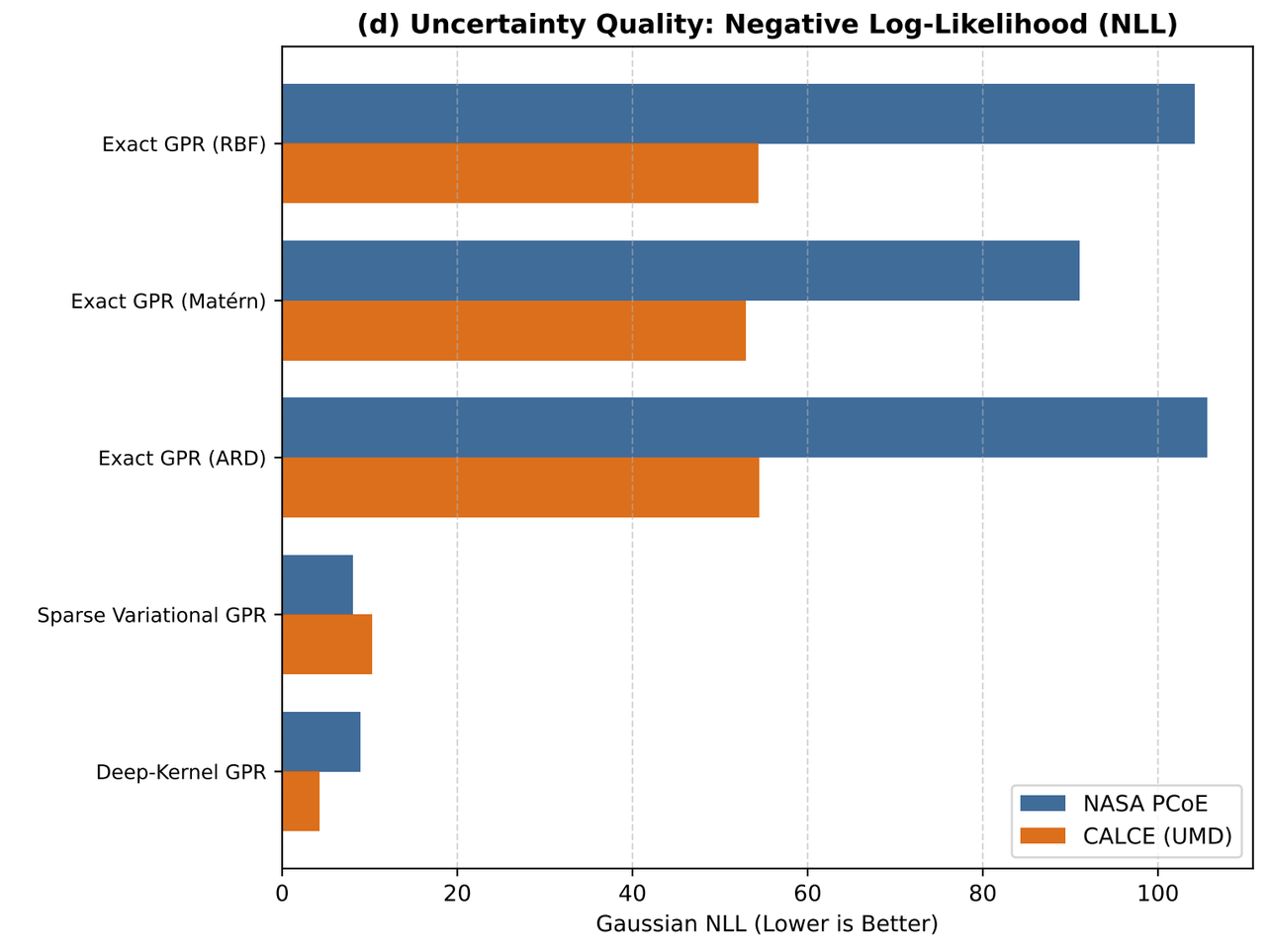

(a) NASA PCoE Benchmark: RMSE (SOH %)
XGBoost
Random Forest
SVR
Deep-Kernel GPR
DeepNN
Wide & Deep NN
WideNN
Sparse Variational GPR
BiLSTM
Exact GPR (Matérn)
Exact GPR (RBF)
Exact GPR (ARD)
Standalone SR
0
20
40
60
80
100
120
140
160
Root Mean Squared Error (%)


(b) CALCE (UMD) Benchmark: RMSE (SOH %)
Deep-Kernel GPR
Random Forest
XGBoost
WideNN
BiLSTM
DeepNN
Wide & Deep NN
Sparse Variational GPR
SVR
Exact GPR (Matérn)
Exact GPR (RBF)
Exact GPR (ARD)
Standalone SR
−200
0
200
400
600
800
1000
Root Mean Squared Error (%)


(c) Generalization Coefficient of Determination (R²)
SVR
Random Forest
XGBoost
Standalone SR
Exact GPR (RBF)
Exact GPR (Matérn)
Exact GPR (ARD)
Sparse Variational GPR
Deep-Kernel GPR
WideNN
DeepNN
Wide & Deep NN
BiLSTM
NASA PCoE
CALCE (UMD)
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
R² Score (Clipped to [-0.2, 1.0])


(d) Uncertainty Quality: Negative Log-Likelihood (NLL)
Exact GPR (RBF)
Exact GPR (Matérn)
Exact GPR (ARD)
Sparse Variational GPR
Deep-Kernel GPR
NASA PCoE
CALCE (UMD)
0
20
40
60
80
100
Gaussian NLL (Lower is Better)